A 50 year old conjecture, disproven **change title**

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Abstract

Will do historical perspective: where did it start, why they cared, build up why they conjectured what they conjectured (reasoning and logic behind it), then a deep dive in the disproof with a focus on visualization and helping build understanding/intuition.

1 AI/Data usage

Used Claude to generate some TikZ's code with prompt:
Give me standalone TikZ code for a Venn-style containment diagram: one large circle labeled 'no-lonely-colour graphs' fully containing a smaller circle labeled 'minimal Cayley graphs', to show the proper-subset relationship. Use plain black outlines, light gray fill on the inner circle, and make sure the outer label doesn't overlap the inner circle. Compile it to check it renders before sending it to me.

2 Introduction

No-lonely-colored graphs have quite strict requirements set upon their structure. Are these constraints sufficient enough to bound the chromatic number of the graph?

3 Prerequisites

The prime question originally asked by Babai is basically: Is the chromatic number of a graph bounded whenever a no-lonely-colored edge coloring exists? Interesting to see that in 1995 Babai actually changed his ideas, he conjectured that: If our graph is no-lonely-colored, we can't say anything about the chromatic number.

3.1 Cayley Graphs

3.2 Minimal Cayley Graphs

A Cayley Graph is minimal if removing any element from the set that generates it, does not generate that cayley graph anymore. (So the generating set is minimal with respect to the Cayley graph). The minimal cayley graph has the least amount of edges while staying a representation of a group generated by S.

In a minimal Cayley graph, **no cycle can contain a color exactly once**, this is forced by the minimality: So basically we assume that we have a minimal cayley graph, and then we assume that we have a lonely edge in a cycle in that graph. Then by showing that that cycle can be rewritten as a product that expresses the lonely edge's generator as a product of the other generators, we see that that generator is redundant and hence the cayley graph was not minimal, giving us a contradiction.

$$\langle S \setminus \{s\} \rangle = \langle S \rangle = \Gamma(G, S)$$

Hence a lonely edge can't appear in a minimal cayley graph.

[**proper proof necessary - Babai 1978**]

3.3 No-lonely-colored graphs

It is quite evident that since all Cayley graphs are no-lonely-colored graphs, that Cayley Graphs are a subset of the broader spectrum of no-lonely-colored graphs.

[**give example of a no-lonely-colored graph that is not Cayley here**]

So if we find/conjecture anything that holds for all Cayley graphs, maybe then it will then also hold for all no-lonely-colored graphs.

3.4 Girth

The girth of a graph is just the length of the shortest possible cycle. Most of the times we use this to ask the question: How does this trait hold given that the length of the shortest possible cycle (girth) is at least some $g \in \mathbb{R}$. You basically forbid a graph from being 'locally dense', hence we force the graph to look more tree-like. (find reference)

3.5 Chromatic number

The chromatic number of a graph is the smallest number of colors needed to color the **vertices** of the graph so that no two adjacent vertices (vertices that are connected by an edge) share the same color.

3.6 Babai's main conjecture

No-lonely-colored graphs have a bounded chromatic number (Babai 1978):
Babai conjectured that there is some link between coloring the edges and coloring the vertices, where if we constrain the coloring of the edges, that also constraints the independent coloring of the vertices. This is quite surprising (for now), since I don't see right away what link there would be between these two concepts. It is evident that some kind of regularity is induced by no-lonely-coloring of graphs, hence it seems most likely the case that Babai's intuition about this regularity gave him the idea that these constraints adjust what possible graphs can be created so much that it induces a bounded chromatic number. Might this have to do with minimal Cayley Graphs?

3.7 Hypergraphs

Generalization of graphs. Instead of edges connecting vertices, we construct blobs around vertices and all those vertices are connected to each other. Hence a hypergraph where each blob/hyperedge is of size 2 (only has 2 vertices) is just a normal graph.

3.8 r-uniform Hypergraphs

A hypergraph is r-uniform if every hyperedge is connected to exactly r vertices.

3.9 Auxiliary Hypergraphs

A proof technique where, wanting to prove something about a certain Graph G and having some difficulties with that, we construct a new hypergraph whose vertices and hyperedges are chosen in such a way that some property of this axuiliary hypergraph is equivalent too or implies the thing we wanted to prove about G. After proving that easier statement for the hypergraph we translate it back to G to prove our initial conjecture on G. Holds a lot of similarity to auxiliary functions in analysis.

3.10 Bridge

A bridge is a specific edge in a graph G where if removing that edge from the graph results in a connected component being split into multiple smaller connected components and hence the total number of connected components increasing.

3.11 Closed walks on a hypergraph

Similar to a closed walk for a graph, but then we walk from vertex to vertex along hyperedges (which, if we recall, connects multiple vertices together), we are allowed to use each hyperedge multiple times, just not right after eachother

3.12 Tranquil Hypergraphs

Take a r-uniform hypergraph, and label all vertices in each hyperedge individually. So a vertex might have multiple different labels, representing different hyperedge's perspectives. If one of these collection of labels exists, where for every closed walk, the projection graph built by hopping from one vertex in a hyperedge to another, and hence connecting the labels of the two endpoint labels of that hop, is bridgeless. Then we call our hypergraph tranquil.

3.13 Monochromatic colorings

This one is quite easy, a coloring is monochromatic if only one color is used.

3.14 Gallai-Witt theorem

Basically, no matter the given integers k, d, no matter how the set $\mathbb{Z}^d$ is colored with k colors, and no matter the finite set chosen $T \subset \mathbb{Z}^d$, we can always find a way to scale and translate T in such a way that the result is a specific subset of points of $\mathbb{Z}^d$ that have the same color.

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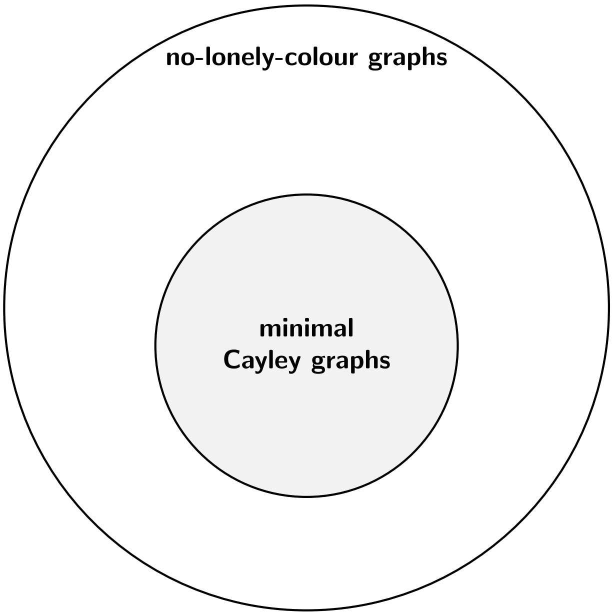
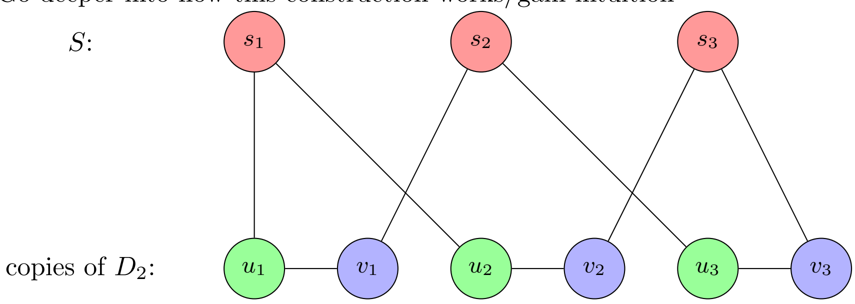


We know from the paper that no-lonely colored graphs do not have a bounded chromatic number, but what about minimal cayley graphs?

4 New concepts

Stable/independent sets, Perfect matchings, Blanche Descartes/Tutte construction
Go deeper into how this construction works/gain intuition



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